

Paramount Diagnostic Lab

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

1 - 2 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Evaluate thyroid function and assess abnormal binding protein disorders

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

Circulating antibodies to thyroid hormones have also been reported to interfere with free hormone immunoassays.

Custom Additional Information

Triiodothyronine (T_3) normally represents only approximately 5% of the thyroid hormone and like thyroxine is almost entirely bound to the carrier proteins, with only 0.25% of the total being in the free state. Measurement of free triiodothyronine is of value in confirming the diagnosis of hyperthyroidism, when an elevated free or total thyroxine level is found. Abnormal total and free triiodothyronine concentrations may appear in T_3 toxicosis, in the presence of normal thyroxine levels. Free T_3 levels are unaffected by carrier protein variation.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.¹

Age	Range (pg/mL)
0 to 3 d	2.0–7.9
4 to 30 d	2.0–5.2
31 d to 12 m	1.6–6.4
13 m to 5 y	2.0–6.0
6 to 10 y	2.7–5.2
11 to 19 y	2.3–5.0
>19 y	2.0–4.4

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; improper labeling

References

Albertini A, Ekins RP, eds. *Free Hormones in Blood*. Amsterdam, Holland: Elsevier Biomedical Press;1982.

Parslow ME, Oddie TH, Fisher DA, et al. Evaluation of serum tri-iodothyronine and adjusted tri-iodothyronine (free tri-iodothyronine index) in pregnancy. *Clin Chem*. 1977 Mar; 23(3):490-492. PubMed 402243 (<http://www.ncbi.nlm.nih.gov/pubmed/402243>)

Footnotes

1. *Reference Intervals for Children and Adults*. Elecsys Thyroid Test. Roche Diagnostics; May 2005. PubMed 8595709 (<http://www.ncbi.nlm.nih.gov/pubmed/8595709>)